INCH-POUND

MIL-PRF-32516/1
04 August 2022

PERFORMANCE SPECIFICATION SHEET

# INTERMITTENT FAULT DIAGNOSTIC EQUIPMENT (ELECTRICAL), DEPOT LEVEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

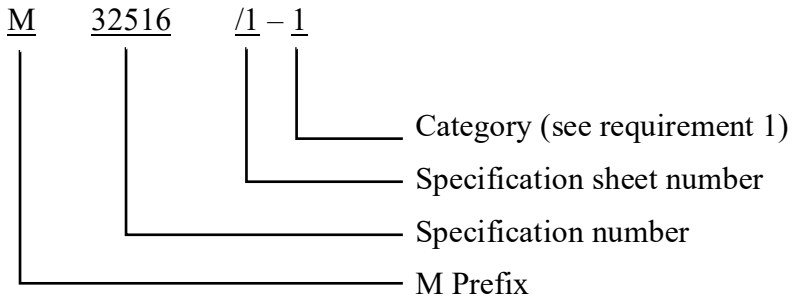
The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-32516.

## REQUIREMENTS

1. Classification. Diagnostic equipment manufactured in accordance with this specification sheet shall be classified as Category 1.

2. Part or Identifying Number (PIN). The PIN to be used for the diagnostic equipment acquired to this specification sheet is created as follows:

Part or Identifying Number (PIN) example:




3. Operating environment. The diagnostic equipment shall be capable of detecting intermittent faults in the operating environment specified in MIL-PRF-32516 for depot level diagnostic equipment. This equipment is not intended to be portable or for use in field level applications.

4. Performance requirements. Depot level diagnostic equipment shall satisfy the performance requirements of MIL-PRF-32516 when subjected to the first article inspection tests of MIL-PRF-32516 (MIL-PRF-28800, class 4).

5. Expandability. The diagnostic equipment shall be expandable up to a minimum of 10,000 test points without loss of fault detection and isolation performance capability. The final number of test points shall be specified and approved by the procuring activity (see MIL-PRF-32516, 6.2.u).



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6. Size and weight. Diagnostic equipment shall be installed in cabinets/racks for ease of installation and maintenance. The preferred cabinet/rack openings to accommodate the diagnostic equipment is 19 inches (482.6 mm). See EIA/ECA-310 for cabinet/rack guidance. The individual diagnostic equipment components shall not exceed the portability individual item weight (see MIL-STD-1472) of 35 pounds (16 kg) unless otherwise specified by the procuring activity.

7. Environmental stimulus capability. The diagnostic equipment shall be capable of interfacing with combined temperature/vibration environmental test facilities. See Note 1. The purpose of the test chamber is to environmentally stimulate the Line Replaceable Unit (LRU)/Weapon Replaceable Assembly (WRA) or weapon system's Electrical Wiring Interconnect System (EWIS) to replicate the environmental fault.

8. Test data display. The diagnostic equipment shall display test program parameters and output data to indicate the number of occurrences and identify the conductive path(s) in which intermittent faults have occurred. The display may be separate from the diagnostic equipment cabinet/rack.

NOTES

1. Intended use. The diagnostic equipment is intended for use in detecting and isolating intermittent faults in LRU/WRA, chassis, and backplanes and their wire harnesses; weapon system EWIS; or patch cables, electronic test cables, and their connectors. The LRU/WRA is typically tested with shop replaceable assemblies removed. The LRU/WRA or EWIS is removed from the weapon system and installed in an environmental test chamber capable of combined temperature/vibration testing to simulate the weapon system operating conditions in which the fault originally occurred. The LRU/WRA or EWIS may be installed in the environmental test chamber for testing prior to weapon system installation or removal from the weapon system for testing/repair. Appendices A through C of MIL-PRF-32516 provide recommended guidelines for defining temperature/vibration weapon system operating conditions for testing LRU/WRA or weapon system EWIS. For LRU/WRA and weapon system, EWIS troubleshooting beyond the detection and isolation of intermittent faults, the designated automatic test equipment (ATE) should be used.

2. Referenced documents. In addition to MIL-PRF-32516 and its applicable specification sheets, this specification sheet references:

| | |
|---|---|
| MIL-PRF-28800 | Test Equipment for Use with Electrical and Electronic Equipment |
| MIL-STD-1472 | Human Engineering |
| EIA/ECA-310 | Cabinets, Racks, Panels, and Associated Equipment |



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CONCLUDING MATERIAL

Custodians:

Army - MI
Navy - AS
Air Force - 85

Preparing activity:

Navy - AS
Project 6625-2021-007

Review activity:
Air Force - 170

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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